SQL Cheat Sheet

Basic Queries

- Select all columns:

```
SELECT * FROM employees;
```

- Select specific columns:

```
SELECT first_name, last_name, salary FROM employees;
```

- Filter by condition:

```
SELECT * FROM employees WHERE salary > 8000;
```

- Filter by multiple conditions:

```
SELECT * FROM employees WHERE department_id = 50 OR department_id = 80;
```

- Filter by pattern matching:

```
SELECT * FROM employees WHERE first_name LIKE 'M%';
```

- Sort results:

```
SELECT first_name, last_name, salary FROM employees ORDER BY salary DESC;
```

- Limit results:

```
SELECT * FROM employees FETCH FIRST 10 ROWS ONLY;
```

String Functions

- Concatenate columns:

```
SELECT first_name || ' ' || last_name AS "Full Name" FROM employees;
```

- Convert to uppercase/lowercase:

```
SELECT UPPER(first_name), LOWER(last_name) FROM employees;
```

- Find string length:

```
SELECT LENGTH(first_name) FROM employees;
```

- Find position of a character:

```
SELECT INSTR(last_name, 'a') FROM employees;
```

- Extract part of a string:

```
SELECT SUBSTR(first_name, 1, 2) FROM employees;
```

Date Functions

- Current date:

```
SELECT SYSDATE FROM dual;
```

- Add months:

```
SELECT ADD_MONTHS(SYSDATE, 3) FROM dual;
```

- Find months between two dates:

```
SELECT MONTHS_BETWEEN(SYSDATE, hire_date) FROM employees;
```

- Get last day of the month:

```
SELECT LAST_DAY(SYSDATE) FROM dual;
```

- Get next occurrence of a weekday:

```
SELECT NEXT_DAY(SYSDATE, 'Monday') FROM dual;
```

- Round and truncate dates:

```
SELECT ROUND(SYSDATE, 'MONTH'), TRUNC(SYSDATE, 'MONTH') FROM dual;
```

Aggregations

- Count rows:

```
SELECT COUNT(*) FROM employees;
```

- Average salary:

```
SELECT AVG(salary) FROM employees;
```

- Sum of salaries:

```
SELECT SUM(salary) FROM employees;
```

- Minimum and maximum salary:

```
SELECT MIN(salary), MAX(salary) FROM employees;
```

Subqueries

- Employees earning more than a specific person:

```
SELECT first_name, last_name, salary FROM employees
WHERE salary > (SELECT salary FROM employees WHERE last_name = 'Davies');
```

- Employees in the same department as someone:

```
SELECT * FROM employees WHERE department_id =
(SELECT department_id FROM employees WHERE last_name = 'Taylor');
```

Joins

- Employees and their department:

```
SELECT first_name, last_name, department_id FROM employees
WHERE department_id IN (SELECT department_id FROM departments);
```

- Employees and their managers:

```
SELECT first_name, last_name, manager_id FROM employees
WHERE manager_id IN (SELECT manager_id FROM employees);
```

Salary Updates

- Increase salary by 10%:

```
SELECT first_name, last_name, salary, salary + salary * 0.10 FROM employees;
```

- Decrease salary by 200:

```
SELECT first_name, last_name, salary, salary - 200 FROM employees;
```